

Global Memory

Global Memory: Memory becoming a burden of AI too?



Mark Li
+852 2123 2645
mark.li@bernsteinsg.com



Edward Hou, CFA
+852 2123 2623
edward.hou@bernsteinsg.com



Yipin Cai, CFA
+852 2123 2669
yipin.cai@bernsteinsg.com

We update our memory & industry models (models for [HBM](#), [DRAM](#), [NAND](#), [Samsung](#), [SK hynix](#), [Micron](#) & [KIOXIA](#)) & maintain Outperform ratings on Samsung, SK hynix & Micron & Underperform on KIOXIA.

HBM price needs to go higher to narrow the profitability gap vs. conventional DRAM.

From 3QCY25 to 2QCY26, conventional DRAM price has risen c. 4.5x, but HBM price is locked by annual contracts and hasn't moved. As the result, we estimate in CY26 deploying capacity to conventional DRAM will generate over 2x revenue & nearly 3x gross profit dollar per wafer capacity vs. HBM. This is why memory suppliers and GPU/XPU companies are negotiating CY27 HBM price now to narrow the gap.

The possible “markup” of GPU/XPU suppliers will amplify the cost burden on hyperscalers. Unlike conventional DRAM & NAND that hyperscalers can source directly from memory suppliers, HBM is packaged in GPUs/XPUs and is part of the COGS of, for example, NVIDIA. NVIDIA needs to mark up the HBM price hike 4x if NVIDIA has 75% gross margin & wants to keep 75% unchanged, despite the HBM price hike. This will make the HBM price hike even a heavier burden for hyperscalers.

With memory making hyperscalers' capex 30% higher, a “re-calibration” is inevitable and may squeeze weaker suppliers. Using Vera Rubin (VR, NVL72) rack as an example, hyperscalers will find their capex higher by 30% if they deploy a data center with this rack and with the same memory capacity installed in the rack, because of higher conventional DRAM & NAND cost, higher HBM cost & the markup charged by NVIDIA. With funding availability and competitive pressure, we think hyperscalers will still invest in AI, but may try to “re-calibrate” their costs across different component suppliers (& possibly even token prices across different customers too?).

To reflect higher HBM price, a notable earnings revision is coming to support the stock prices of Samsung, SK hynix Micron but not KIOXIA. We raise HBM price 2-2.5x, not as much as the size of conventional DRAM price hike, as we assume memory suppliers recognize the strategic value of HBM and choose to price less aggressively. We update our forecast & find our FY27 EPS 25-40% above consensus as the result. As HBM negotiation gradually concludes in the next few months, we expect consensus to move upward and to provide an upward support to the stock of Samsung, SK hynix and Micron, but not KIOXIA as KIOXIA has only NAND & does not have HBM. Please note this also means higher HBM exposure will result lower profitability for memory suppliers.

MediaTek may benefit if hyperscalers prefer to source HBM directly to avoid the possible markup. The stock went up c. 130% in the past 2 months but we reiterate Outperform.

Reiterate Outperform on Samsung, SK hynix and Micron too. We raise target price to KRW440,000 for Samsung (6.2x 1-year forward P/E), KRW3,300,00 for SK hynix (6.2x 1-year forward P/E), and US\$1,300 for Micron (7.7x 1-year forward P/E). Maintain Underperform on KIOXIA (see [KIOXIA: All parties come to an end - Underperform](#)).

BERNSTEIN TICKER TABLE

Ticker	Rating	19 Jun 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
005930.KS (SEC- Samsung)	O	KRW	350,500	440,000	442.0%	KRW	6,611.53	48,393	77,273	53.0	7.2	4.5
OLD				225,000				35,740	49,548			
005935.KS (SEC-Pref - Samsung)	O	KRW	222,000	374,000	308.8%	KRW	6,611.53	48,393	77,273	33.6	4.6	2.9
OLD				191,250				35,740	49,548			
SMSN.LI (Samsung)	O	USD	5,710.00	7,350.00	387.0%	USD	116.15	812.39	1,290.80	49.2	7.0	4.4
OLD				3,888.00				617.62	856.24			
000660.KS (SK hynix)	O	KRW	2,755,000	3,300,000	925.0%	KRW	60,341	395,677	568,862	45.7	7.0	4.8
OLD				1,150,000				286,732	385,594			
MU (Micron)	O	USD	1,133.99	1,300.00	805.2%	USD	8.29	67.39	158.99	136.9	16.8	7.1
OLD				510.00				62.53	121.03			
285A.JP (KIOXIA)	U	JPY	108,600	40,000	4607.8%	JPY	1,014.00	10,013	9,656.84	107.1	10.8	11.2
2454.TT (MediaTek)	O	TWD	4,390.00	4,380.00	204.2%	TWD	66.17	69.52	170.50	66.3	63.1	25.7
ASIA			2,048.07									
EM			1,898.05									
SPX			7,500.58									

PRICE TARGET CHANGE / ESTIMATE CHANGE IN BOLD

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

MU estimate is Adjusted EPS; MU valuation is Adjusted P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Samsung, SK hynix, Micron: We maintain **Outperform** on Samsung, SK hynix and Micron and raise price targets to **KRW440,000 (vs 225,000)**, **KRW3,300,000 (vs 1,150,000)** and **US\$1,300 (vs 510)** respectively, driven by increased forecasts of conventional memory pricing as well as HBM pricing going forward. We now value them using 1-year forward P/E and target **6.2x** for Samsung, **6.2x** for SK hynix and **7.7x** for Micron.

DETAILS

Please feel free to download our industry ([DRAM](#), [NAND](#), [CoWoW/HBM](#)) and company models ([Samsung](#), [SK hynix](#), [Micron](#) & [KIOXIA](#)).

We think HBM price will be higher. And added with the "markup" on HBM, the increase will make AI feel the burden of memory price hike too. A "re-calibration" to share the cost burden is probably inevitable.

- As a result of the DRAM supply shortage, and skyrocketing conventional DRAM prices, we think HBM price is also bound to go up next year, by 2-2.5x vs this year ([Exhibit 1-Exhibit 2](#)), and will exert a heavy pressure on server rack cost and also hyperscaler capex. The cost pressure possibly won't be limited to just the higher purchase cost of HBM stacks paid to memory suppliers. Take NVIDIA as an example. HBM is packaged inside NVIDIA's GPUs, and the HBM cost paid memory suppliers is part of NVIDIA's COGS. Simply passing through this HBM cost increase to NVIDIA's customers will dilute NVIDIA's gross margin. Assume NVIDIA earns 75% gross margin on the VR200 racks, NVIDIA will need to increase its rack price by 4x the cost increase of HBM in order to maintain margin at 75%.
- We estimate before the HBM price increase HBM was 5% of VR200 price that NVIDIA charges. With the HBM price increase alone, the selling price of VR200 needs to be higher by 6% price increase. If NVIDIA marks up by 4x to keep gross margin at 75%, the rack price can go up by 24%. For a hyperscaler deploying VR200 racks, its total data center capex (including out-of-rack capex) needs to go up by 4% if only HBM cost is passed on, or 15% if all markup is included. Additionally, conventional DRAM and NAND prices have both gone up substantially too, and contribute another 14% increase to the total data center capex ([Exhibit 3](#)). All together with 2-2.5x HBM price increase+markup, and conventional DRAM & NAND price increase (& no markup), we find hyperscalers' AI capex needs to be higher by 30% just to cover higher memory costs. Should they build their ROI analysis based on the original budget of VR200, they likely now need to redo the analysis with "I" (investment) 30% higher than original.
- In summary, the burden of memory price hike is now propagating from consumer applications to AI too. Hyperscalers need to redo their ROI analysis, as memory is now making the investment requirement bigger. Given the competitive pressure and funding availability, we don't think this will make hyperscalers slow down AI investments. A "re-calibration" is however inevitable, so that hyperscalers, supply chain & maybe even token price together make adjustments to cope with the burden. Weaker suppliers may be squeezed in this "re-calibration" but companies in our coverage are well positioned and some may emerge as a beneficiary of this "re-calibration".
- Please note that the analysis here isn't perfect and the estimate is simply a ballpark one. First it is very possible that hyperscalers will mix VR200 servers with those with other accelerators and/or with CPUs only for different computation and agentic needs. Further some conventional DRAM and NAND will be pre-installed in the servers shipped by NVIDIA and hence may be subject to markups. The analysis above however for simplistic sake assumes no markups - essentially assuming NVIDIA allows customers to weigh truer computation needs against memory cost and all conventional DRAM & NAND are installed by customers. Finally, GPU & accelerator providers may argue the existence of the markups on HBM. They will highlight HBM being an integral part of the entire system, and the integration of HBM with logic dies, package, and other components in the rack & software. They may hence argue that they charge no markups on HBM, & treat HBM just as "pass-through" revenue but deserve high margin on the rest of the system. In any case, it is true the HBM is part of the COGS of these GPUs/accelerators and the gross margin % of these GPU/accelerators suppliers will contract if they simply pass the higher HBM cost downward without a markup. They will be tempted to charge a markup if they would like to defend the gross margin %, but that as discussed above resulting in an amplified capex burden on hyperscalers.

EXHIBIT 1: We forecast 2-2.5x increase in HBM prices YoY next year so that the profitability of HBM & conventional DRAM can narrow.

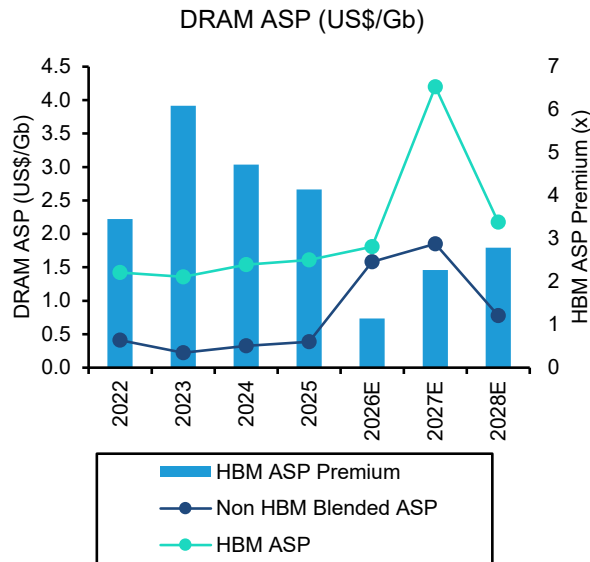
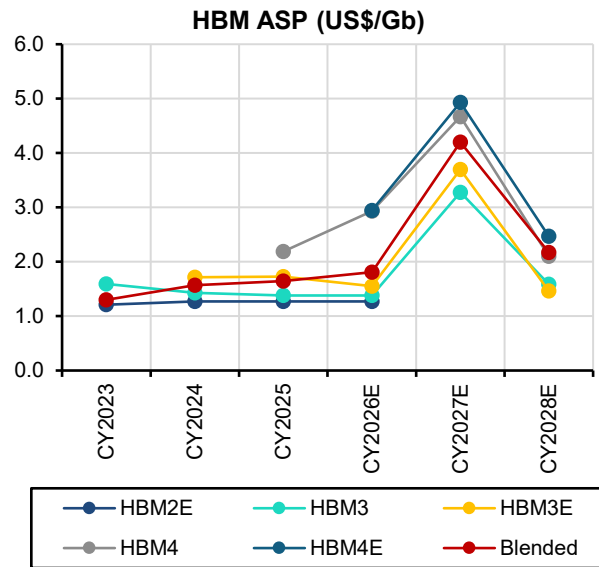
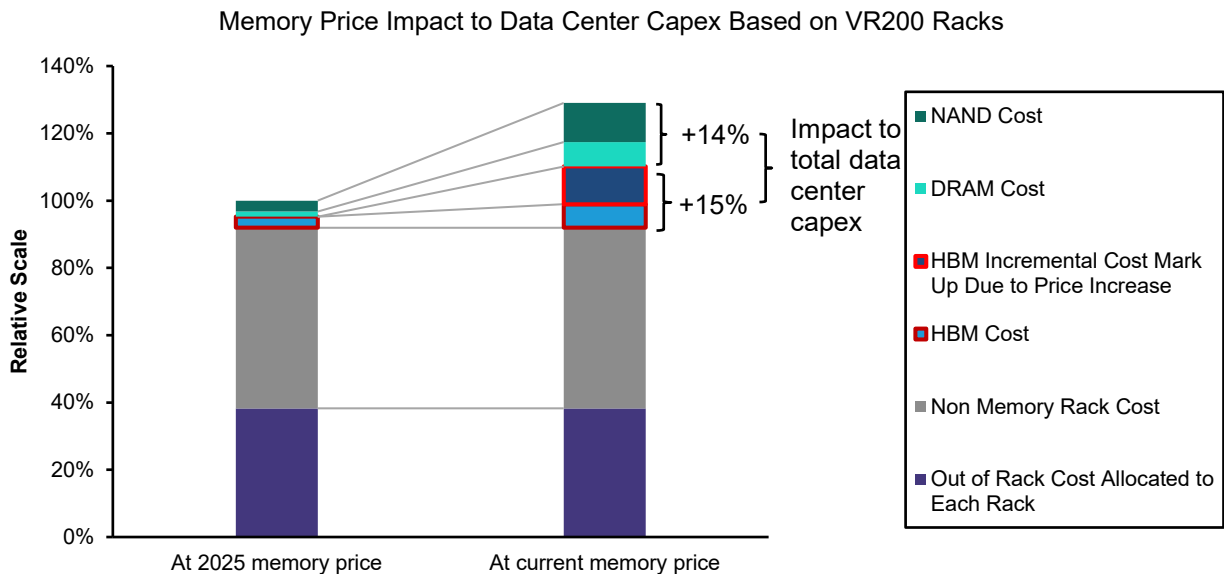


EXHIBIT 2: We expect the price increase to take place at all generations of HBM.



Source: Company reports, TrendForce, Bernstein estimates and analysis

EXHIBIT 3: We estimate HBM price increase, its cost markup, and also DRAM & NAND price increase together require nearly 30% increase in AI data center capex.



Source: Expert conversations, company reports, TrendForce, Bernstein estimates and analysis

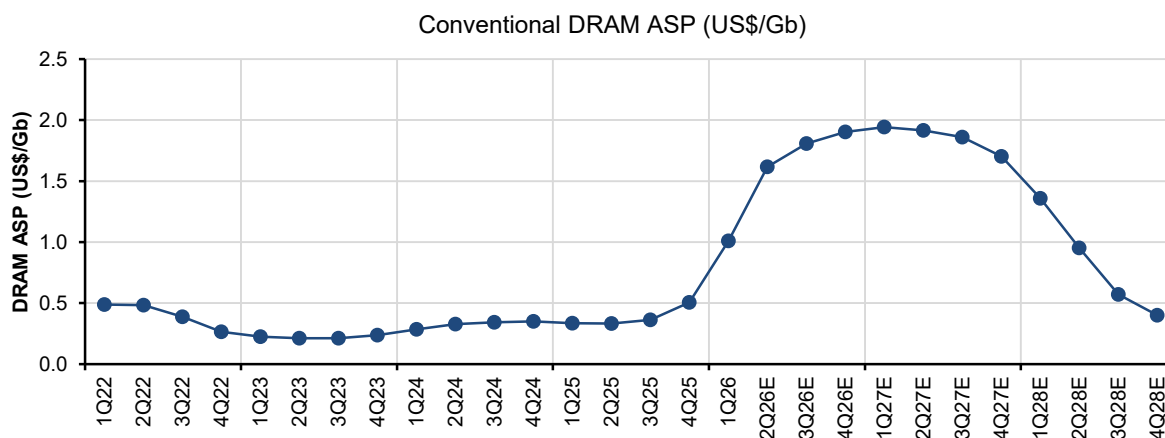
The profitability of HBM may stay below that of conventional memory next year should suppliers recognize the strategic value of HBM and price HBM less aggressively.

- Conventional DRAM price is expected to have moved up by ~4.5x by 2QCY26 compared to 3QCY25, and we think prices can possibly rise by another turn in CY27 (or another ~25% from 2QCY26) before peaking ([Exhibit 4-Exhibit 5](#)). As a result of that, conventional DRAM now enjoys similar if not higher ASP per bit than HBM. And given its higher bit density and yield, conventional DRAM, we estimate, will generate 2x revenue per wafer capacity vs. HBM this year ([Exhibit 6](#)), and command a

notably higher margins too ([Exhibit 7](#)). Both Samsung and Micron clearly said non-HBM DRAM margin was already ahead of HBM margin in their 1QCY26 earnings calls, as conventional DRAM price keeps climbing, the margin gap between the two keeps widening.

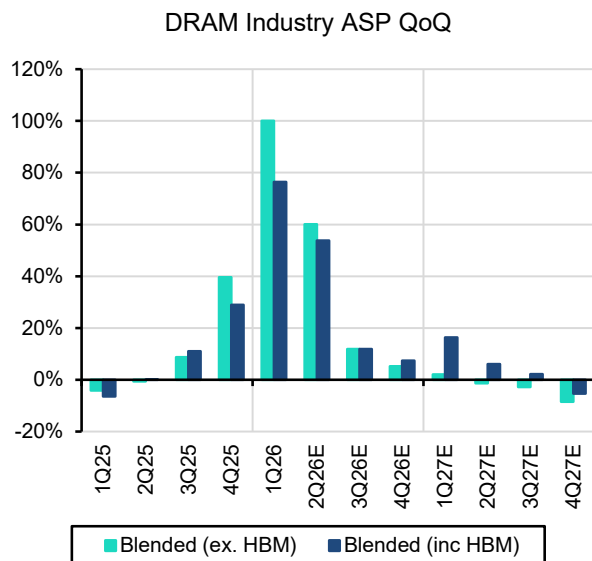
- Given conventional DRAM price may rise further next year, we think memory suppliers will demand a significant increase in HBM prices to balance the profitability between the two. Samsung in its earnings call also mentioned that it expects the margin differential to be “significantly reduced in 2027”, implying raising HBM prices. We estimate 3x increase in HBM price is needed for HBM revenue per wafer capacity to catch up with conventional DRAM. Some, on the other hand, may argue HBM is a vital part of AI and is critical to the entire AI infrastructure build out. We think suppliers may recognize this and understand that high HBM cost can be unhealthy for overall AI ecosystem development and ultimately to memory demand as well. SK hynix indeed also stressed that it will try to achieve “optimal allocation between HBM and general DRAM”, and will not try to maximize revenue. We hence model 2-2.5x increase in HBM price next year on account of the above considerations ([Exhibit 1-Exhibit 2](#)). HBM profitability hence will remain below that of conventional DRAM next year, but the gap should be much smaller than this year.
- Across different HBM suppliers, Samsung we believe now leads in HBM4 technology. Our latest Korea memory export tracker (report [link](#)) did find a meaningful increase in export value per weight from Samsung in May, which can be due to HBM4 starting shipment ([Exhibit 8](#)). We think this can enable Samsung to grow its HBM market share this year and next year ([Exhibit 9-Exhibit 10](#)). However, as mentioned above, more HBM exposure may likely mean less profit for memory suppliers. And we do get the impression that Samsung is more focused on achieving higher profit margins, and hence may choose to deploy more capacity for conventional memory, even with Samsung's technological lead in HBM4.

EXHIBIT 4: Conventional DRAM price has gone up by 4.5x and we see another 25% increase before reaching the peak.



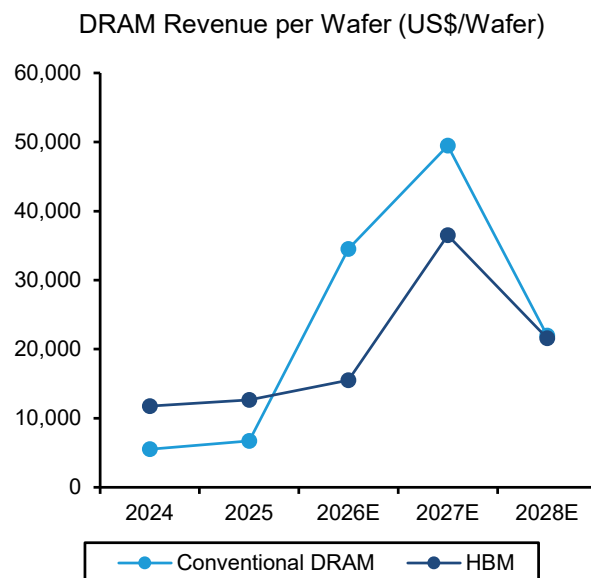
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 5: We forecast DRAM price to continue rising after 2QCY26 though at a slower rate and reach peak in CY27.



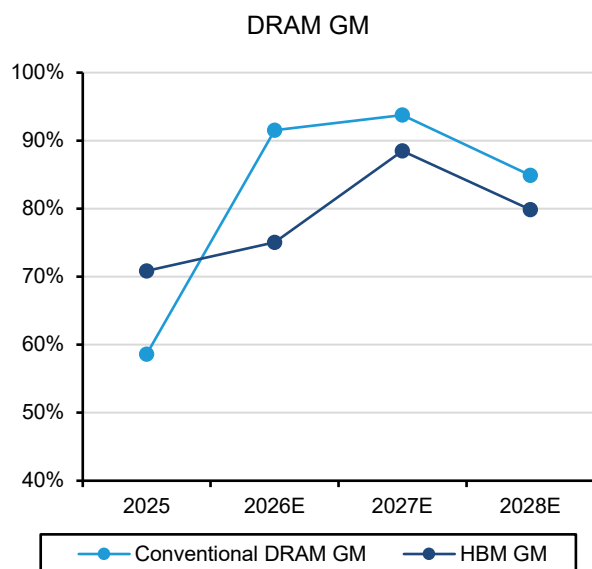
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 6: Conventional DRAM now generates considerably more revenue per wafer capacity than HBM, and we project an HBM price hike next year to narrow that gap.



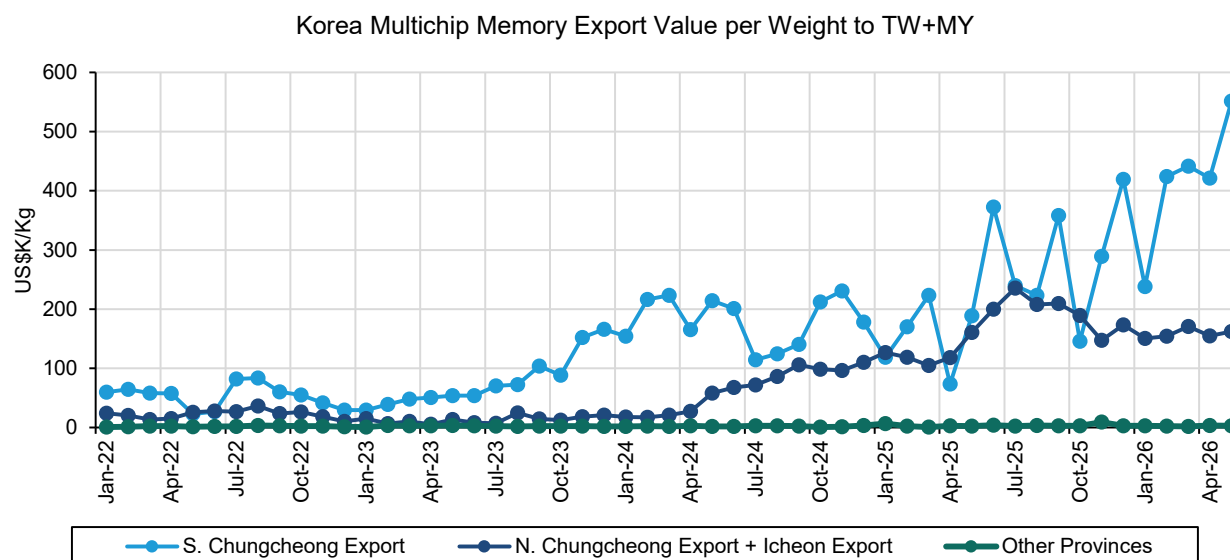
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 7: We believe the margin gap between HBM and conventional DRAM to narrow next year.



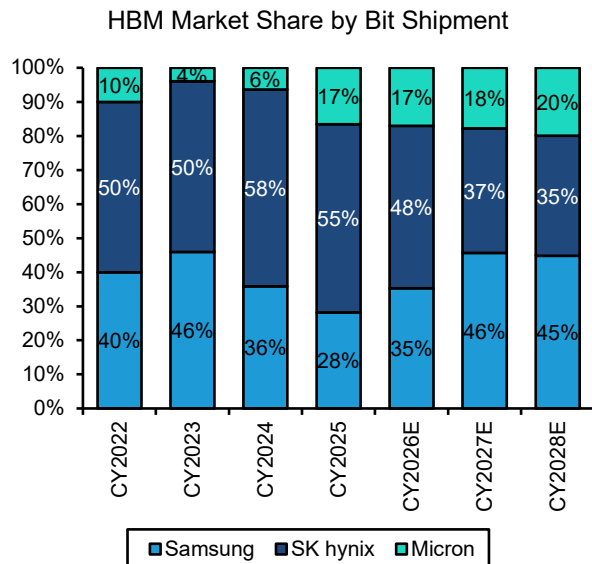
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 8: Value per weight remained largely in the same range though the export value per weight for Samsung rose notably & suggested HBM4 shipment in May.



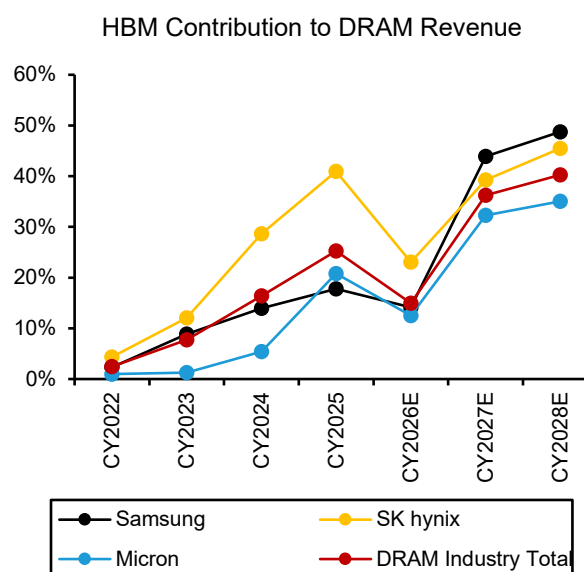
Source: Korea Custom Service, KITA and Bernstein analysis

EXHIBIT 9: We see Samsung to gain share in HBM on better HBM4 performance & more capacity, but also see uncertainty as Samsung may choose to allocate more capacity to conventional DRAM for better profitability.



Source: TrendForce, Bernstein estimates and analysis

EXHIBIT 10: With the expected HBM price increase, we forecast HBM's revenue mix in DRAM to rebound next year.



Source: TrendForce, company reports, Bernstein estimates and analysis

DRAM stocks will benefit from a rapid upward earnings revision in the near term.

- We believe negotiation of HBM supply for next year is still ongoing and we should hear more news on that in the coming months. And we also don't think this HBM price increase has been fully reflected in the sell-side consensus on Bloomberg. We will elaborate our model revision below, but we're above consensus in CY2027 meaningfully as we incorporate HBM price hike in this revision. We expect other sell-side forecasts to follow and hence a notable upward change to CY2027

consensus soon, which in turn support stock prices to move upward in the near term.

- Please note this HBM revision will only benefit Samsung, SK hynix and Micron. Pure NAND suppliers, such as KIOXIA, do not have HBM and hence will not benefit from this revision.

MediaTek may benefit if hyperscalers demand to source HBM directly to avoid the markup.

- Though we expect NVIDIA & other AI accelerator suppliers to emphasize the importance of integrating HBM with the rest of their offerings and hence continuing having HBM as part of their revenue, many hyperscalers, we believe, will negotiate and try to source HBM directly to avoid the chance of HBM markup. We believe this trend to favor Asian ASIC service providers as their business model allows hyperscalers to do so now.
- MediaTek thus can benefit from this trend. The execution on its first and second TPU project is also solid. Our recent supply chain checks also suggest an upside risk to our 2028 projection. The stock went up ~130% in the past 2 months, but we still see upside and reiterate Outperform.

We raise our projections on stronger conventional DRAM prices and also higher HBM pricing. Based on 6.2-7.7x 1 year forward P/E, we raise target price for Samsung, SK hynix and Micron and see another 15-26% upside. Maintain Outperform.

- We raise forecasts for Samsung, SK hynix and Micron to reflect stronger-than-expected price increase in the near term, but more importantly our CY2027 projections as a result of higher HBM ASP as discussed above. And we are now also notably above consensus in FY27 EPS by 25-38% ([Exhibit 11-Exhibit 13](#)). Earnings are forecast to rise almost vertically this year and into next year as well ([Exhibit 14-Exhibit 16](#)), and that will keep the stock prices strong.
- We generally see the DRAM shortage to be more acute and remain for longer than we previously expected. We see DRAM prices can remain strong likely for most of CY27. That said we do still believe with more clean rooms coming online, prices will likely soften in CY28. And we hence still model revenue to decline YoY in FY28 for all three companies, that puts us below consensus, which apparently assumes very limited decrease in pricing. Please note that our forecast still has DRAM industry gross margin at 70% exiting CY28, higher than peak margins in all past upcycles only except for CY18 ([Exhibit 17](#)).
- Valuing memory companies using P/B is tricky now as margins and return on equity will be reaching unprecedented levels ([Exhibit 18](#)). Hence we cannot reference history to identify the right P/B multiple. We also note that cash position will also reach an unprecedented level, also making it complicated to value based on book value ([Exhibit 19-Exhibit 24](#)). We hence switch to P/E based valuation, and we believe investors are also increasingly basing their valuation on earnings. We set our target P/E close to historical trough levels to value peak cycle earnings. We raise target price to KRW440,000 for Samsung, at 6.2x 1 year forward P/E, representing 26% upside. SK hynix target price is raised to KRW 3,300,000 at 6.2x forward P/E as well, representing 20% upside. Micron target price is set at US\$1,300, at 7.7x P/E, representing 15% upside. We reiterate **Outperform** ratings on Samsung, SK hynix and Micron.

EXHIBIT 11: For Samsung, our EPS is 26% above consensus for 2027.

	2026E					2027E					2028E				
	Bernstein					Bernstein					Bernstein				
Samsung	New	Old	% Chg	Consensus	% Diff	New	Old	% Chg	Consensus	% Diff	New	Old	% Chg	Consensus	% Diff
Revenue (KRW B)	678,209	571,178	18.7%	684,036	-0.9%	920,481	686,889	34.0%	862,952	6.7%	591,825	446,595	32.5%	917,793	-35.5%
YoY	103.3%	71.2%		105.0%		35.7%	20.3%		26.2%		-35.7%	-35.0%		6.4%	
Gross Margin	77.8%	81.2%	-3.4%	69.4%	8.4%	88.6%	87.3%	1.2%	73.3%	15.2%	66.9%	52.9%	14.0%	68.2%	-1.3%
Operating Margin	64.1%	57.2%	6.8%	51.2%	12.9%	76.1%	65.4%	10.7%	56.0%	20.1%	45.4%	32.9%	12.4%	54.2%	-8.8%
EPS (KRW)	48,393	35,740	35.4%	44,557	8.6%	77,273	49,548	56.0%	61,510	25.6%	31,054	17,340	79.1%	63,146	-50.8%
YoY	632.0%	440.6%		573.9%		59.7%	38.6%		38.0%		-59.8%	-65.0%		2.7%	

Source: Bloomberg, Bernstein estimates and analysis

EXHIBIT 12: We raise estimates for hynix and see 30%+ upside to consensus for both 2026 and 2027.

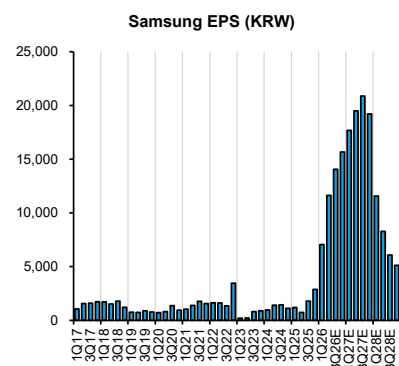
SK hynix	2026E					2027E					2028E				
	Bernstein New	Old	% Chg	Consensus	% Diff	Bernstein New	Old	% Chg	Consensus	% Diff	Bernstein New	Old	% Chg	Consensus	% Diff
Revenue (KRW Bn)	369,627	307,701	20.1%	339,075	9.0%	582,611	406,986	43.2%	491,637	18.5%	319,374	182,106	75.4%	542,332	-41.1%
YoY	280.5%	216.7%		249.0%		57.6%	32.3%		45.0%		-45.2%	-55.3%		10.3%	
Gross Margin	87.9%	87.0%	0.9%	83.5%	4.4%	91.7%	88.4%	3.3%	82.6%	9.1%	81.2%	70.7%	10.6%	79.4%	1.8%
Operating Margin	83.1%	81.1%	2.0%	76.7%	6.3%	84.8%	82.4%	2.4%	77.7%	7.0%	64.0%	59.5%	4.5%	74.9%	-10.9%
Diluted EPS (KRW)	395,677	286,732	38.0%	300,629	31.6%	568,862	385,594	47.5%	429,536	32.4%	235,417	124,299	89.4%	466,745	-49.6%
YoY	555.7%	375.2%		398.2%		43.8%	34.5%		42.9%		-58.6%	-67.8%		8.7%	

Source: Bloomberg, Bernstein estimates and analysis

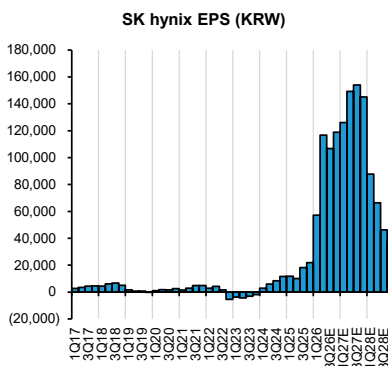
EXHIBIT 13: Our Micron FY27 EPS forecast is 38% above consensus too.

Micron	FY2026E					FY2027E					FY2028E				
	Bernstein New	Old	% Chg	Consensus	% Diff	Bernstein New	Old	% Chg	Consensus	% Diff	Bernstein New	Old	% Chg	Consensus	% Diff
GAAP															
Revenue (US\$B)	122.6	116.9	4.8%	114.1	7.5%	253.8	199.8	27.0%	196.0	29.5%	207.1	110.9	86.8%	205.5	0.8%
YoY	228.0%	212.8%		205.1%		107.0%	70.9%		71.8%		-18.4%	-44.5%		4.9%	
Gross Margin	78.9%	78.0%	1.0%	77.6%	1.4%	87.8%	85.5%	2.3%	81.8%	5.9%	82.7%	70.3%	12.4%	78.7%	4.1%
Operating Margin	73.6%	71.9%	1.7%	71.9%	1.7%	84.2%	80.7%	3.4%	80.1%	4.1%	77.8%	61.5%	16.3%	76.3%	1.5%
Diluted EPS (US\$)	66.99	62.27	7.6%	61.28	9.3%	158.96	121.31	31.0%	112.35	41.5%	121.42	52.54	131.1%	119.25	1.8%
YoY	782.7%	720.6%		707.6%		137.3%	94.8%		83.3%		-23.6%	-56.7%		6.1%	
Non-GAAP															
Diluted EPS (US\$)	67.39	62.53	7.8%	61.82	9.0%	158.99	121.03	31.4%	115.19	38.0%	121.40	52.75	130.1%	117.02	3.7%
YoY	713.3%	654.6%		646.1%		135.9%	93.6%		86.3%		-23.6%	-56.4%		1.6%	

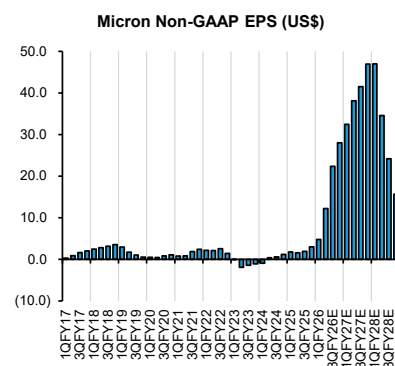
Source: Bloomberg, Bernstein estimates and analysis

EXHIBIT 14: We expect EPS to grow exponentially going forward.

Source: Company reports, Bernstein estimates and analysis

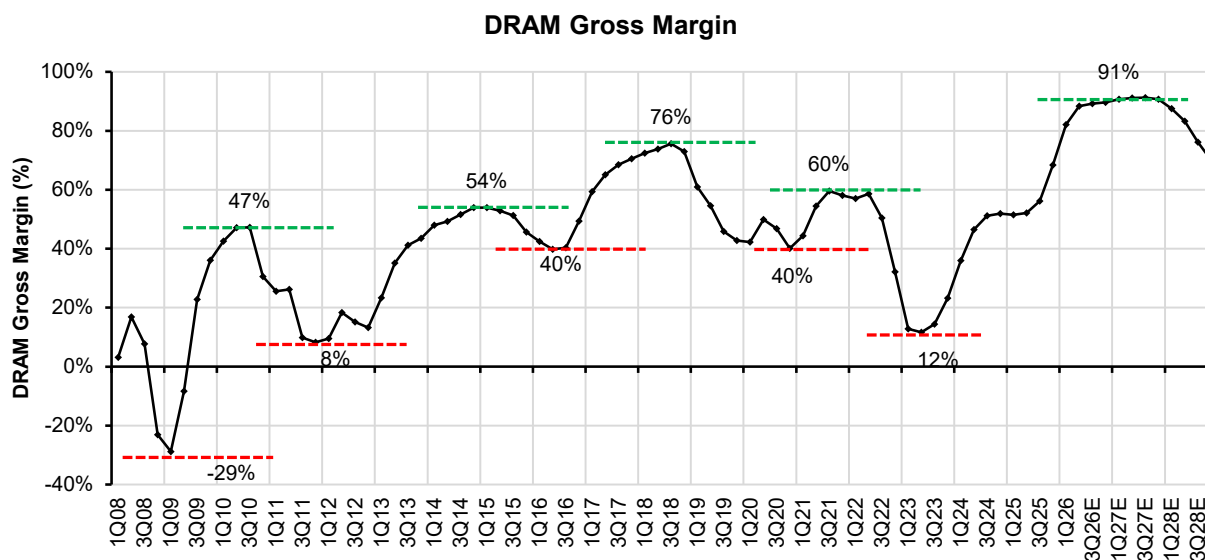
EXHIBIT 15: We model earnings to reach peak in 2H CY27...

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 16: ... and then gradually decline in CY28 as prices normalize.

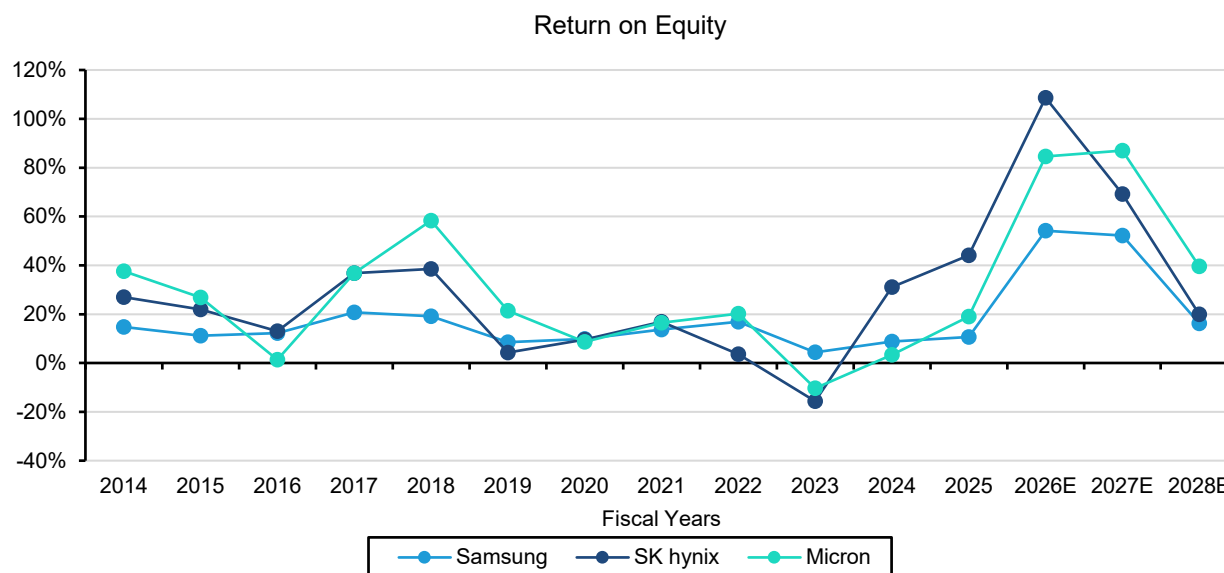
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 17: **We forecast DRAM prices to fall in CY28 but GM remain at 70% exiting CY28, higher than peak margins in most of past upcycles.**

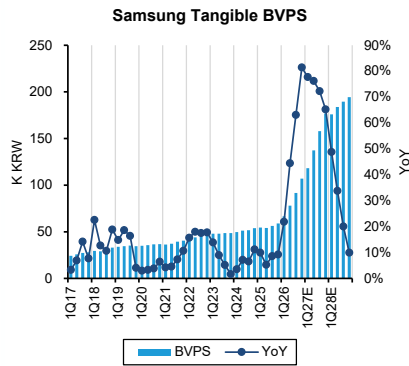
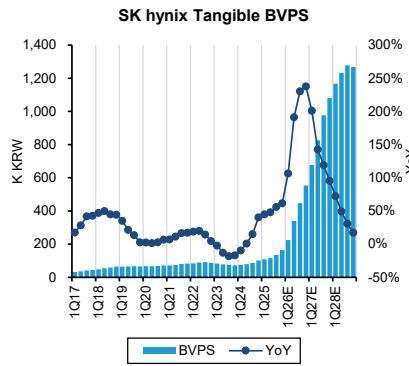
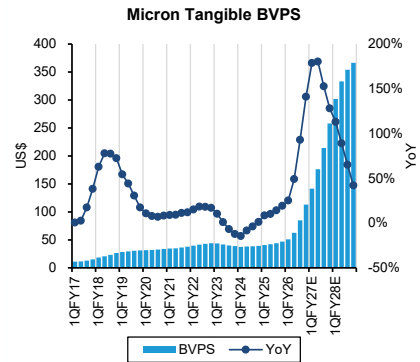
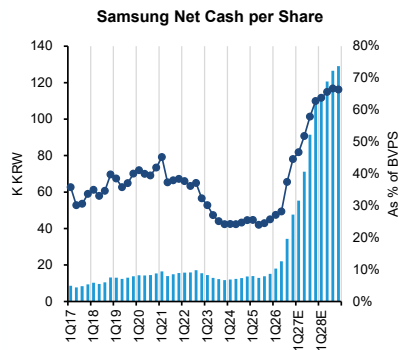
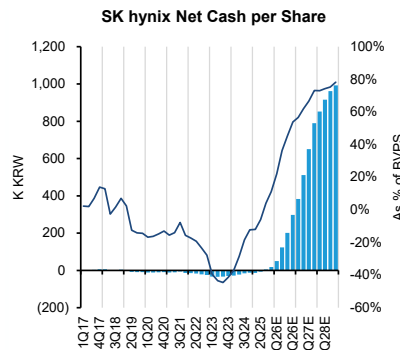
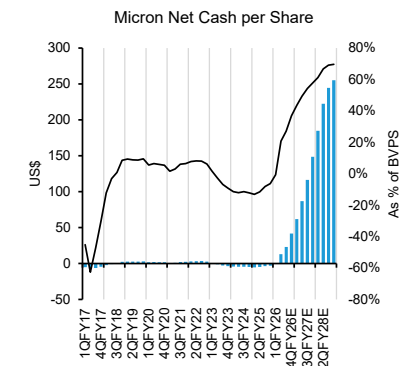
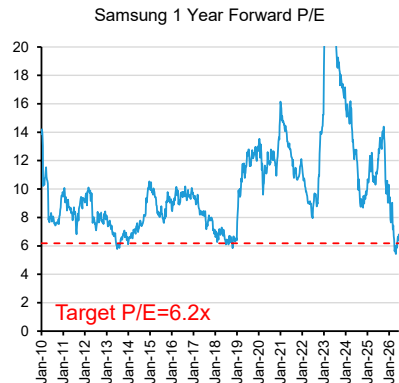
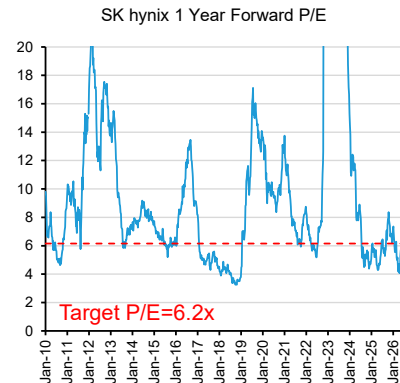
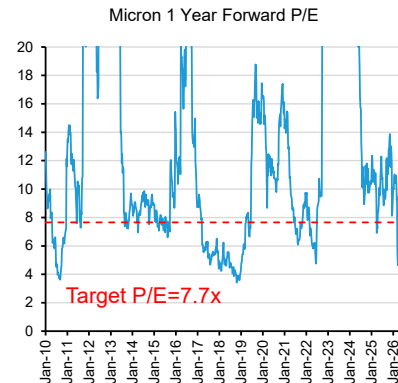


Source: Company reports, DRAMeXchange, Bernstein estimates and analysis

EXHIBIT 18: **ROE will rise to unprecedented levels.**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 19: Samsung BVPS can grow by nearly 2x in both 2026 and 2027.**EXHIBIT 20: SK hynix can see even steeper accretion in BVPS.****EXHIBIT 21: We see 4x increase in Micron BVPS from now to end of FY27.****EXHIBIT 22: Cash will accumulate at unprecedented pace as well.****EXHIBIT 23: We forecast cash to reach 70-80% of book value...****EXHIBIT 24: ... unless company meaningful increase payout or use it for M&A.****EXHIBIT 25: We now target 6.2x 1 year P/E to value Samsung.****EXHIBIT 26: We also use 6.2x P/E for hynix.****EXHIBIT 27: Micron target P/E is 7.7x, still close to past trough levels.**

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 28: Samsung Income Statement

Figures in KRW B Unless Otherwise Stated

Income Statement	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Revenue	333,606	678,209	920,481	591,825	79,141	74,566	86,062	93,837	133,873	161,466	185,767	197,102
COGS	202,236	150,586	105,386	195,676	51,010	49,070	52,570	49,586	51,960	32,727	35,122	30,776
Gross Profit	131,370	527,623	815,095	396,149	28,131	25,497	33,492	44,251	81,913	128,739	150,645	166,326
R&D	37,740	40,168	46,024	41,428	9,033	9,019	8,824	10,864	11,337	9,688	9,288	9,855
SG&A	50,029	53,062	68,808	86,151	12,413	11,801	12,502	13,313	13,344	12,917	13,004	13,797
Operating Profit	43,601	434,393	700,262	268,570	6,685	4,676	12,166	20,074	57,233	106,134	128,353	142,674
Other non-operating Income/(Expenses)	—	—	—	—	865	18	(124)	(101)	245	—	—	—
Equity-Method Investments	671	638	706	745	119	159	304	90	121	175	172	171
Net interest (expense)/gain	4,550	3,773	9,283	16,540	1,483	903	1,200	965	1,230	613	747	1,183
Income Before Taxes	48,823	438,804	710,251	285,855	9,152	5,756	13,546	21,028	58,828	106,921	129,272	144,028
Income Taxes	4,275	116,164	195,319	78,610	929	640	1,320	1,387	11,603	29,403	35,550	39,608
Net Income (before minority interest)	44,548	322,640	514,932	207,245	8,223	5,116	12,226	19,642	47,225	77,518	93,722	104,420
Minority Interest	287	245	480	500	194	182	219	350	124	66	144	156
Net Income (after minority interest)	44,261	322,395	514,452	206,745	8,028	4,934	12,007	19,292	47,101	77,452	93,578	104,264
EPS After Minority (KRW per share)	6,612	48,393	77,273	31,054	1,192	737	1,801	2,886	7,056	11,634	14,056	15,661
Total Shares Out	6,695	6,662	6,658	6,658	6,737	6,692	6,665	6,684	6,675	6,658	6,658	6,658
EBITDA	88,008	482,161	757,985	336,392	17,415	16,033	23,126	31,434	68,713	117,747	140,457	155,244
Margins %												
Gross Margin	39.4%	77.8%	88.6%	66.9%	35.5%	34.2%	38.9%	47.2%	61.2%	79.7%	81.1%	84.4%
Operating Margin	13.1%	64.1%	76.1%	45.4%	8.4%	6.3%	14.1%	21.4%	42.8%	65.7%	69.1%	72.4%
NI before MI Margin	13.4%	47.6%	55.9%	35.0%	10.4%	6.9%	14.2%	20.9%	35.3%	48.0%	50.5%	53.0%
NI after MI Margin	13.3%	47.5%	55.9%	34.9%	10.1%	6.6%	14.0%	20.6%	35.2%	48.0%	50.4%	52.9%
EBITDA Margin	26.4%	71.1%	82.3%	56.8%	22.0%	21.5%	26.9%	33.5%	51.3%	72.9%	75.6%	78.8%
ROE %	10.6%	54.2%	52.1%	16.2%	8.1%	5.1%	12.0%	18.5%	40.9%	59.2%	61.7%	59.3%
Growth %												
Revenue Growth	10.9%	103.3%	35.7%	(35.7%)	4.4%	(5.8%)	15.4%	9.0%	42.7%	20.6%	15.1%	6.1%
Operating Profit Growth	33.2%	896.3%	61.2%	(61.6%)	3.0%	(30.1%)	160.2%	65.0%	185.1%	85.4%	20.9%	11.2%
EPS Growth	33.6%	632.0%	59.7%	(59.8%)	6.8%	(38.1%)	144.3%	60.2%	144.5%	64.9%	20.8%	11.4%
Division Financials												
Semiconductor	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Total Revenue	130,123	542,942	858,436	443,457	25,131	27,875	33,117	44,000	81,700	133,040	155,527	172,675
DRAM	72,143	360,236	622,242	328,287	12,814	13,813	18,210	27,306	54,571	90,735	103,086	111,844
NAND	31,871	153,560	202,715	76,708	6,261	7,362	8,466	9,782	20,214	35,085	45,372	52,889
Total Memory	104,100	513,856	825,017	405,055	19,100	21,200	26,700	37,100	74,800	125,835	148,474	164,747
System LSI	25,823	28,886	33,219	38,202	5,981	6,625	6,367	6,850	6,850	7,155	7,003	7,878
Operating Profit	24,847	420,139	685,688	246,129	1,106	350	6,991	16,400	53,700	103,159	124,292	138,989
OP margin %	19.1%	77.4%	79.9%	55.5%	4.4%	1.3%	21.1%	37.3%	65.7%	77.5%	79.9%	80.5%
Display Panel	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Total Revenue	29,848	30,707	32,625	31,903	5,867	6,380	8,102	9,500	6,700	6,970	8,260	8,777
LCD	1,913	2,287	2,328	2,371	411	447	486	570	536	536	604	611
AMOLED	27,935	28,420	30,297	29,532	5,456	5,933	7,615	8,930	6,164	6,434	7,657	8,166
Operating Profit	4,160	3,495	4,062	4,233	462	473	1,225	2,000	400	558	1,219	1,319
OP margin %	13.9%	11.4%	12.5%	13.3%	7.9%	7.4%	15.1%	21.1%	6.0%	8.0%	14.8%	15.0%
IT & Mobile	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Total Revenue	129,600	132,554	132,076	133,444	37,000	29,200	34,100	29,300	38,100	31,118	33,643	29,692
Mobile (handsets + tablets)	104,882	107,681	107,958	109,327	30,754	23,735	27,496	22,897	31,124	24,221	27,916	24,420
Other (PCs, smartwatches, VR etc.)	21,618	21,618	20,537	20,537	5,446	4,765	6,004	5,403	6,376	6,057	4,846	4,339
Networking	3,100	3,255	3,581	3,581	800	700	600	1,000	600	840	882	933
Operating Profit	12,900	9,111	9,194	15,545	4,300	3,100	3,600	1,900	2,800	2,047	2,343	1,920
OP margin %	10.0%	6.9%	7.0%	11.6%	11.6%	10.6%	10.6%	6.5%	7.3%	6.6%	7.0%	6.5%
Consumer Electronics	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Total Revenue	57,300	56,410	53,983	53,303	14,500	14,100	13,900	14,800	14,300	14,012	13,827	14,271
Visual Display	30,900	31,308	30,136	30,649	7,800	7,000	7,300	8,800	7,700	7,480	7,557	8,571
Appliance	26,400	25,102	23,847	22,655	6,700	7,100	6,600	6,000	6,600	6,532	6,270	5,700
Operating Profit	(200)	339	(130)	938	300	200	(100)	(600)	200	70	69	-
OP Margin %	(0.3%)	0.6%	(0.2%)	1.8%	2.1%	1.4%	(0.7%)	(4.1%)	1.4%	0.5%	0.5%	-
Harman	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Total Revenue	15,157	16,711	18,382	20,220	3,419	3,830	3,954	3,954	3,800	4,213	4,349	4,349
Operating Profit	1,511	1,376	1,448	1,725	307	484	421	300	200	300	429	446
OP Margin %	10.0%	8.2%	7.9%	8.5%	9.0%	12.6%	10.6%	7.6%	5.3%	7.1%	9.9%	10.3%

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 29: Samsung Operating Metrics

Division Operating Metrics												
Semiconductor	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
DRAM ASP (\$/Gb)	0.46	1.84	2.76	1.17	0.38	0.38	0.44	0.62	1.19	1.86	2.06	2.21
% change	20.9%	297.5%	49.7%	(57.5%)	(19.0%)	(0.2%)	15.8%	40.0%	92.6%	56.7%	10.3%	7.7%
NAND ASP (\$/GB)	0.07	0.27	0.30	0.09	0.07	0.06	0.07	0.08	0.16	0.26	0.32	0.34
% change	(7.0%)	289.1%	8.8%	(68.7%)	(16.0%)	(4.0%)	5.6%	24.0%	88.5%	66.0%	22.0%	7.0%
DRAM Bits Shipped (Gb)	109,195	131,034	150,689	186,854	23,107	25,880	29,762	30,446	31,268	32,519	33,495	33,752
% change	9.7%	20.0%	15.0%	24.0%	2.2%	12.0%	15.0%	2.3%	2.7%	4.0%	3.0%	0.8%
NAND Bits Shipped (GB)	321,027	378,812	458,363	554,619	65,378	83,030	91,333	81,286	88,196	90,401	95,825	104,391
% change	1.8%	18.0%	21.0%	21.0%	(11.0%)	27.0%	10.0%	(11.0%)	8.5%	2.5%	6.0%	8.9%
DRAM Cost/Gb	0.19	0.20	0.25	0.22	0.19	0.19	0.20	0.20	0.19	0.19	0.21	0.22
% change	6.6%	4.9%	22.3%	(10.0%)	(11.3%)	(1.5%)	5.1%	(0.7%)	(5.0%)	2.9%	8.0%	8.0%
NAND Cost/GB	0.05	0.05	0.04	0.04	0.06	0.06	0.05	0.05	0.05	0.05	0.05	0.05
% change	(2.8%)	(7.8%)	(13.4%)	(11.4%)	4.1%	(2.5%)	(10.1%)	(5.4%)	1.9%	(0.0%)	1.0%	(3.0%)
NAND gross margins	25.8%	82.4%	86.0%	60.3%	14.0%	12.6%	25.6%	43.3%	69.3%	81.5%	84.7%	86.1%
DRAM gross margins	58.5%	89.1%	91.0%	81.1%	50.2%	50.9%	55.4%	68.4%	84.4%	89.8%	90.0%	89.9%
Display Panel	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
LCD Panel Units (mn)	1.8	2.0	2.2	2.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0.5
% change	25.0%	10.0%	10.0%	10.0%	(10.0%)	5.0%	7.0%	10.8%	(10.0%)	—	15.0%	3.3%
AMOLED Panel Units (mn)	351.2	328.9	345.9	381.1	71.1	83.0	94.1	103.0	79.6	79.8	83.3	86.3
% change	2.4%	(6.3%)	5.2%	10.2%	(26.9%)	16.7%	13.3%	9.4%	(22.7%)	0.3%	4.4%	3.5%
LCD Panel ASP (USD)	748.2	776.9	715.5	662.3	695.2	746.6	767.1	775.3	801.7	785.7	770.0	754.6
% change	17.4%	3.8%	(7.9%)	(7.4%)	7.7%	7.4%	2.8%	1.1%	3.4%	(2.0%)	(2.0%)	(2.0%)
AMOLED ASP (USD)	55.9	58.0	58.5	51.8	52.8	51.0	58.4	59.7	52.8	53.9	61.4	63.3
% change	(6.4%)	3.8%	0.9%	(11.5%)	(7.3%)	(3.4%)	14.4%	2.4%	(11.6%)	2.0%	14.0%	3.0%
IT & Mobile	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Smartphone Units (mn)	240	213	212	232	61	58	61	60	57	51	54	50
% change	7.1%	(11.4%)	(0.5%)	9.4%	17.3%	(4.9%)	5.2%	(1.6%)	(5.0%)	(10.0%)	6.0%	(7.9%)
Featurephone Units (mn)	15	15	15	15	4	4	4	4	4	4	4	4
% change	—	—	—	—	—	—	—	—	—	—	—	—
Total Handset Units (mn)	255	228	227	247	65	62	65	64	61	55	58	54
% change	6.7%	(10.7%)	(0.4%)	8.8%	16.1%	(4.6%)	4.9%	(1.5%)	(4.7%)	(9.4%)	5.6%	(7.4%)
Total Tablet Units (mn)	27	27	27	28	7	7	7	6	7	7	7	6
% change	(3.3%)	0.6%	1.0%	1.0%	0.5%	(0.5%)	—	(14.3%)	16.7%	1.0%	—	(14.3%)
Smartphone ASP (USD)	286.3	316.2	316.8	293.4	326.0	270.0	304.0	244.0	349.5	290.1	319.1	303.1
% change	(2.3%)	10.4%	0.2%	(7.4%)	25.4%	(17.2%)	12.6%	(19.7%)	43.2%	(17.0%)	10.0%	(5.0%)
Featurephone ASP (USD)	43.3	45.2	43.5	41.7	42.5	43.1	42.9	44.7	45.9	45.5	45.0	44.6
% change	(2.9%)	4.4%	(3.9%)	(3.9%)	(3.9%)	1.2%	(0.4%)	4.2%	2.7%	(1.0%)	(1.0%)	(1.0%)
Total Handset ASP (USD)	271.9	298.1	298.5	277.9	309.4	256.0	288.7	232.1	330.5	273.2	301.2	284.9
% change	(2.0%)	9.7%	0.1%	(6.9%)	26.1%	(17.2%)	12.7%	(19.6%)	42.4%	(17.3%)	10.2%	(5.4%)
Total Tablet ASP (USD)	159.7	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8
% change	(0.1%)	0.1%	0.0%	—	—	—	—	—	—	—	—	—
Consumer Electronics	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
TV Unit Shipments (K)	41,341	39,354	37,106	37,279	10,142	9,331	10,077	11,791	9,904	9,409	9,409	10,632
% change	(0.0%)	(4.8%)	(5.7%)	0.5%	(13.0%)	(8.0%)	8.0%	17.0%	(16.0%)	(5.0%)	—	13.0%
TV ASP (USD)	498	506	511	516	503	506	494	491	502	502	507	512
% change	(4.4%)	1.6%	1.0%	1.0%	0.7%	0.5%	(2.3%)	(0.7%)	2.3%	—	1.0%	1.0%

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 30: Samsung Balance Sheet and Cash Flow Statement

Figures in KRW B Unless Otherwise Stated

Balance Sheet	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Cash & Cash Equivalents	125,847	340,909	759,221	878,584	105,134	100,728	108,464	125,847	147,378	176,196	256,904	340,909
Accounts Receivable	51,128	113,679	140,106	76,548	44,867	43,551	50,539	51,128	82,285	97,192	110,945	113,679
Inventory	48,030	37,031	24,610	53,809	53,220	51,037	50,332	48,030	58,278	50,911	43,850	37,031
Other Current Assets	22,680	23,009	25,773	12,944	19,465	16,845	20,106	22,680	18,279	22,046	22,828	23,009
Total Current Assets	247,685	514,629	949,710	1,021,884	222,686	212,161	229,441	247,685	306,220	346,345	434,527	514,629
Net PP&E	215,305	253,537	297,152	350,292	207,386	205,026	204,861	215,305	217,815	227,016	235,739	253,537
Intangibles	29,481	29,644	29,644	29,644	26,695	25,998	26,469	29,481	29,644	29,644	29,644	29,644
Investments	48,030	55,137	55,842	56,587	33,489	33,240	38,345	48,030	54,620	54,794	54,966	55,137
Other Non-Current Assets	26,442	36,868	43,908	23,447	26,121	28,451	24,543	26,442	25,041	30,203	34,748	36,868
Total Non-Current Assets	319,258	375,186	426,546	459,970	293,691	292,715	294,219	319,258	327,120	341,656	355,097	375,186
Total Assets	566,942	889,815	1,376,257	1,481,854	516,377	504,875	523,660	566,942	633,340	688,002	789,624	889,815
Accounts Payable	67,113	37,850	39,287	65,371	63,433	55,598	57,731	67,113	67,825	39,269	39,726	37,850
Total Debt	25,239	28,139	28,139	28,139	11,144	14,030	16,672	25,239	28,139	28,139	28,139	28,139
Other Liabilities	38,270	68,883	88,756	50,825	35,186	35,686	35,755	38,270	50,740	58,893	68,786	68,883
Total Liabilities	130,622	134,872	156,182	144,334	109,762	105,313	110,158	130,622	146,704	126,301	136,651	134,872
Shareholders' Equity	436,320	754,943	1,220,075	1,337,520	406,614	399,562	413,502	436,320	486,636	561,701	652,973	754,943
Total Liabilities & Equity	566,942	889,815	1,376,257	1,481,854	516,377	504,875	523,660	566,942	633,340	688,002	789,624	889,815
Cash Flow Statement	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Net Income	45,207	322,885	514,932	207,245	8,223	5,116	12,226	19,642	47,225	77,518	93,722	104,420
Depreciation & Amortization	46,062	49,423	59,378	69,477	11,144	11,771	11,374	11,774	11,894	12,028	12,518	12,983
Changes in Net Working Capital & Other	(5,909)	(54,149)	(3,205)	55,058	(2,787)	473	(980)	(2,616)	(18,849)	(33,292)	(1,840)	(167)
Cash Flow From Operations	85,360	318,159	571,105	331,780	16,580	17,360	22,620	28,800	40,270	56,253	104,400	117,236
Capex	(47,525)	(89,140)	(101,338)	(120,961)	(12,130)	(13,035)	(10,810)	(11,550)	(17,130)	(20,815)	(20,828)	(30,367)
Other	(15,687)	(1,342)	(1,655)	(1,655)	(1,440)	(4,707)	(2,231)	(7,310)	(100)	(414)	(414)	(414)
Cash Flow From Investing	(63,212)	(90,482)	(102,993)	(122,617)	(13,570)	(17,742)	(13,040)	(18,860)	(17,230)	(21,228)	(21,242)	(30,781)
Cash Flow From Financing	(13,477)	(16,567)	(49,800)	(89,800)	(11,340)	(3,407)	(4,190)	5,460	(5,460)	(6,207)	(2,450)	(2,450)
Free Cash Flow	37,835	229,019	469,767	210,818	4,450	4,325	11,811	17,250	23,140	35,439	83,572	86,869
BVPS	58,952	106,942	176,734	194,300	54,600	53,988	56,197	58,952	66,523	77,961	91,649	106,942

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 31: SK hynix Income Statement

Figures in KRW B Unless Otherwise Stated

Income Statement	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Revenue	97,147	369,627	582,611	319,374	17,639	22,232	24,449	32,827	52,576	90,679	107,699	118,672
COGS	38,457	44,749	48,497	59,921	7,537	10,249	10,420	10,251	10,897	11,138	11,268	11,446
Gross Profit	58,690	324,878	534,114	259,452	10,102	11,983	14,029	22,576	41,679	79,541	96,431	107,227
SG&A	5,585	10,060	30,389	45,268	1,189	1,294	1,170	1,931	2,593	2,267	2,585	2,615
R&D	5,899	7,786	9,733	9,733	1,472	1,476	1,476	1,476	1,476	1,900	2,000	2,411
Other Operating Exp. (Inc.)	(1)	(0)	0	-	-	-	-	(1)	-	-	-	-
Operating Income	47,207	307,032	493,993	204,451	7,441	9,213	11,383	19,170	37,610	75,374	91,846	102,201
Net Interest Expense	(430)	(38)	(54)	(40)	(152)	(129)	(103)	(46)	22	(21)	(20)	(19)
Other Inc/(Exp)	3,688	38,985	-	-	2,010	(361)	3,510	(1,471)	13,985	25,000	-	-
Earnings Before Tax	50,465	345,978	493,939	204,411	9,299	8,723	14,790	17,653	51,617	100,353	91,826	102,182
Income Tax/(Benefit)	7,517	64,256	88,909	36,794	1,191	1,726	2,193	2,407	11,271	18,064	16,529	18,393
Net Income	42,948	281,722	405,030	167,617	8,108	6,997	12,597	15,246	40,346	82,290	75,298	83,789
Basic EPS (KRW)	62,108	399,606	574,511	237,755	11,756	10,135	18,242	21,852	57,175	116,723	106,805	118,850
Diluted EPS (KRW)	60,341	395,677	568,862	235,417	11,411	9,572	17,850	21,522	56,670	115,575	105,755	117,681
Dividend per share (KRW)	3,000	11,500	41,633	48,647	375	375	375	1,875	375	375	375	10,375
Basic WASO	692	705	705	705	690	690	690	696	705	705	705	705
Diluted WASO	712	712	712	712	711	712	712	712	712	712	712	712
EBITDA	61,138	322,871	515,519	231,574	10,787	12,667	14,943	22,741	41,339	79,149	95,959	106,424
Margins												
Gross Margin	60.4%	87.9%	91.7%	81.2%	57.3%	53.9%	57.4%	68.8%	79.3%	87.7%	89.5%	90.4%
EBITDA Margin	62.9%	87.4%	88.5%	72.5%	61.2%	57.0%	61.1%	69.3%	78.6%	87.3%	89.1%	89.7%
Operating Margin	48.6%	83.1%	84.8%	64.0%	42.2%	41.4%	46.6%	58.4%	71.5%	83.1%	85.3%	86.1%
Net Income Margin	44.2%	76.2%	69.5%	52.5%	46.0%	31.5%	51.5%	46.4%	76.7%	90.7%	69.9%	70.6%
ROE	44.1%	108.7%	69.1%	19.9%	41.8%	33.2%	53.8%	55.3%	113.2%	160.3%	106.1%	93.2%
Growth												
Revenue Growth	46.8%	280.5%	57.6%	(45.2%)	(10.8%)	26.0%	10.0%	34.3%	60.2%	72.5%	18.8%	10.2%
Operating Profit Growth	101.2%	550.4%	60.9%	(58.6%)	(7.9%)	23.8%	23.6%	68.4%	96.2%	100.4%	21.9%	11.3%
EPS Growth	116.1%	543.4%	43.8%	(58.6%)	1.2%	(13.8%)	80.0%	19.8%	161.6%	104.2%	(8.5%)	11.3%
Key Operating Metrics	2025	2026E	2027E	2028E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Total Revenue	97,147	369,627	582,611	319,374	17,639	22,232	24,449	32,827	52,576	90,679	107,699	118,672
DRAM Revenue (KRW B)	75,496	273,776	456,206	270,975	14,041	17,208	19,168	25,080	41,220	68,419	79,026	85,111
Gb Shipped (M Gb)	98,961	117,763	136,605	172,122	20,062	24,877	26,743	27,278	27,142	29,313	30,485	30,823
ASP (\$/Gb)	0.54	1.56	2.23	1.05	0.48	0.49	0.52	0.63	1.04	1.56	1.73	1.85
Cost (\$/Gb)	0.16	0.14	0.13	0.14	0.16	0.17	0.17	0.15	0.16	0.14	0.14	0.14
Gb Shipped Change	22.9%	19.0%	16.0%	26.0%	(7.0)%	24.0%	7.5%	2.0%	(0.5)%	8.0%	4.0%	1.1%
ASP Change	31.1%	190.7%	43.2%	(52.9)%	0.0%	2.5%	4.7%	22.5%	63.5%	50.7%	11.1%	6.5%
Cost Change	(7.7)%	(10.6)%	(6.1)%	6.3%	(9.5)%	5.0%	(1.0)%	(12.0)%	8.0%	(10.0)%	(3.0)%	(2.0)%
DRAM Gross Margin	70.2%	90.8%	94.0%	86.4%	66.8%	66.0%	67.9%	76.9%	84.8%	90.9%	92.0%	92.7%
MU DRAM GM												
NAND Revenue (KRW B)	20,536	94,751	125,360	47,287	3,175	4,758	4,988	7,616	11,251	21,911	28,335	33,254
GB Shipped (M GB)	174,408	214,522	263,862	319,273	28,730	48,984	46,045	50,650	45,585	52,422	55,568	60,948
ASP (\$/GB)	0.08	0.30	0.32	0.10	0.08	0.07	0.08	0.10	0.17	0.28	0.34	0.36
Cost (\$/GB)	0.06	0.06	0.05	0.05	0.06	0.06	0.06	0.06	0.07	0.06	0.06	0.06
GB Shipped Change	9.9%	23.0%	23.0%	21.0%	(18.0)%	70.5%	(6.0)%	10.0%	(10.0)%	15.0%	6.0%	9.7%
ASP Change	(5.8)%	257.7%	7.3%	(68.8)%	(20.0)%	(8.9)%	12.7%	32.6%	62.5%	66.0%	22.0%	7.0%
Cost Change	(11.5)%	(3.7)%	(13.1)%	(9.0)%	(14.0)%	(1.8)%	3.5%	(6.8)%	14.1%	(12.0)%	(4.0)%	(4.0)%
NAND Gross Margin	25.1%	79.8%	83.7%	52.3%	17.3%	10.9%	18.1%	42.5%	59.6%	78.6%	83.1%	84.9%
Operating Profit (KRW B)	47,207	307,032	493,993	204,451	7,441	9,213	11,383	19,170	37,610	75,374	91,846	102,201

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 32: SK hynix Balance Sheet and Cash Flow Statement

Figures in KRW B Unless Otherwise Stated

Balance Sheet	2025	2026E	2027E	2028E
Cash & Cash Equivalents	34,942	226,828	565,815	708,105
Accounts Receivable	18,199	75,885	88,649	35,965
Inventories	14,289	14,656	15,083	21,945
Other Current Assets	2,028	2,394	2,394	2,394
Total Current Assets	69,458	319,762	671,941	768,410
PP&E	77,503	106,812	143,952	186,720
Other Non-Current Assets	29,147	34,277	34,300	34,335
Non-Current Assets	106,650	141,089	178,252	221,055
Total Assets	176,108	460,851	850,194	989,465
Debt	22,248	16,768	8,768	8,768
Accounts Payable	2,848	2,787	3,207	4,213
Other Liabilities	30,345	43,383	64,627	69,572
Total Liabilities	55,441	62,938	76,602	82,553
Minority Interest/Other	-	-	-	-
Shareholder's Equity	120,667	397,913	773,592	906,913
Total Liabilities & Equity	176,108	460,851	850,194	989,465
Tangible BVPS	163,847	553,169	1,080,775	1,267,974
Cash Flow Statement	2025	2026E	2027E	2028E
Net Income	42,948	281,722	405,030	167,617
Depreciation & Amortization	13,931	15,839	21,527	27,123
Other	(3,289)	(57,110)	(12,772)	46,828
CF From Operations	53,590	240,452	413,785	241,568
Capex	(27,520)	(44,528)	(58,667)	(69,891)
Other	(3,691)	423	(23)	(35)
CF From Investing	(31,211)	(44,104)	(58,690)	(69,926)
CF From Financing	(1,445)	(7,599)	(16,108)	(29,352)
Cash at Beginning of Period	14,156	34,942	226,828	565,815
Net Cash Flow	20,786	191,887	338,987	142,291
Adjustments	-	(1)	-	-
Cash at End of Period	34,942	226,828	565,815	708,105
Free Cash Flow	26,070	195,924	355,118	171,677

1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
14,311	16,962	27,854	34,942	54,330	105,616	159,596	226,828
10,629	13,125	14,312	18,199	33,808	57,165	71,079	75,885
14,551	13,408	13,156	14,289	15,974	15,498	15,015	14,656
1,737	1,708	1,818	2,028	2,394	2,394	2,394	2,394
41,228	45,203	57,140	69,458	106,506	180,673	248,084	319,762
63,015	64,473	68,145	77,503	82,052	89,410	95,983	106,812
19,742	19,412	23,150	29,147	34,271	34,273	34,275	34,277
82,757	83,885	91,295	106,650	116,323	123,682	130,258	141,089
123,985	129,088	148,435	176,108	222,829	304,355	378,342	460,851
23,333	21,841	24,079	22,248	19,318	18,468	17,618	16,768
1,875	1,806	2,270	2,848	2,798	3,149	2,953	2,787
17,339	18,298	22,080	30,345	36,333	36,333	36,333	43,383
42,547	41,945	48,429	55,441	58,449	57,950	56,904	62,938
-	-	-	-	-	-	-	-
81,439	87,142	100,006	120,667	164,380	246,405	321,438	397,913
123,986	129,087	148,435	176,108	222,829	304,355	378,342	460,851
108,968	116,850	134,904	163,789	225,181	340,383	445,764	553,169
1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
8,108	6,997	12,597	15,246	40,346	82,290	75,298	83,789
3,346	3,454	3,560	3,571	3,729	3,774	4,113	4,223
(2,379)	(1,243)	(1,789)	2,122	(17,645)	(21,226)	(13,627)	(4,612)
9,075	9,208	14,368	20,939	26,430	64,839	65,784	83,400
(6,284)	(4,332)	(5,033)	(11,871)	(7,657)	(11,132)	(10,687)	(15,052)
(3,186)	3	(121)	(387)	429	(2)	(2)	(2)
(9,470)	(4,329)	(5,154)	(12,258)	(7,228)	(11,134)	(10,689)	(15,054)
509	(1,780)	1,375	(1,549)	(2,951)	(2,419)	(1,114)	(1,114)
14,156	14,311	16,962	27,854	34,942	54,330	105,616	159,596
155	2,652	10,892	7,087	19,389	51,286	53,981	67,231
-	(1)	-	1	(1)	-	-	-
14,311	16,962	27,854	34,942	54,330	105,616	159,596	226,828
2,791	4,876	9,335	9,068	18,773	53,707	55,097	68,348

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 33: Micron Income Statement

Figures in USD M Unless Otherwise Stated; Fiscal Year ends August 30

Income Statement (GAAP)	FY2025	FY2026E	FY2027E	FY2028E	1QFY26	2QFY26	3QFY26E	4QFY26E	1QFY27E	2QFY27E	3QFY27E	4QFY27E
Revenue	37,378	122,598	253,776	207,112	13,643	23,860	38,183	46,911	53,529	61,154	65,829	73,264
COGS	22,505	25,829	31,079	35,730	5,997	6,105	6,620	7,107	7,450	7,526	7,864	8,239
Gross Profit	14,873	96,769	222,696	171,381	7,646	17,755	31,563	39,804	46,079	53,627	57,965	65,025
SG&A	1,205	1,567	2,663	2,530	337	344	382	504	589	673	658	743
R&D	3,798	4,937	6,419	7,702	1,171	1,250	1,234	1,282	1,605	1,605	1,605	1,605
Operating Profit (excl. restruct. & Other)	9,870	90,265	213,614	161,149	6,138	16,161	29,947	38,019	43,885	51,350	55,702	62,677
Net Interest Expense	19	290	858	2,205	65	123	37	65	118	173	243	324
Other Non-Operating/Operating Income	(235)	(266)	-	-	(142)	(124)	-	-	-	-	-	-
Earnings Before Tax	9,654	90,288	214,472	163,354	6,061	16,160	29,984	38,084	44,003	51,523	55,945	63,001
Income Tax	1,124	13,821	34,316	27,770	829	2,371	4,528	6,093	7,040	8,244	8,951	10,080
Equity Method Gains/(Losses)	9	4	1	5	8	(4)	(1)	1	1	(0)	-	-
Minority Interest	-	-	-	-	-	-	-	-	-	-	-	-
Net Income (GAAP)	8,539	76,471	180,157	135,589	5,240	13,785	25,455	31,991	36,963	43,279	46,993	52,921
Non-GAAP adjustments	931	1,015	1,033	1,022	242	236	290	248	255	260	262	256
Net Income (Non-GAAP)	9,470	77,487	181,190	136,610	5,482	14,021	25,745	32,239	37,218	43,539	47,256	53,177
Basic EPS (USD)	7.65	67.89	161.24	123.18	4.66	12.24	22.58	28.39	32.87	38.66	42.15	47.66
Diluted GAAP EPS (USD)	7.59	66.99	158.96	121.42	4.60	12.07	22.26	27.99	32.41	38.11	41.55	46.98
Diluted Non-GAAP EPS (USD)	8.29	67.39	158.99	121.40	4.78	12.20	22.38	28.04	32.44	38.11	41.53	46.91
Tangible BVPS (US\$)	47.9	114.0	259.0	367.4	50.8	62.8	84.8	113.0	141.7	176.2	214.3	258.0
Weighted Average Basic Shares (M)	1,116	1,126	1,117	1,101	1,125	1,126	1,127	1,127	1,124	1,120	1,115	1,110
GAAP Diluted Shares Out (M)	1,125	1,142	1,133	1,117	1,138	1,142	1,143	1,143	1,140	1,136	1,131	1,126
Non-GAAP Diluted Shares Out (M)	1,143	1,149	1,140	1,124	1,148	1,149	1,150	1,150	1,147	1,143	1,138	1,133
EBITDA	18,222	100,020	226,977	179,738	8,350	18,447	32,448	40,775	46,836	54,503	59,160	66,478
Margins %												
GAAP Gross Margin	39.8%	78.9%	87.8%	82.7%	56.0%	74.4%	82.7%	84.9%	86.1%	87.7%	88.1%	88.8%
GAAP EBITDA Margin (excl. restruct. & Other)	48.8%	81.6%	89.4%	86.8%	61.2%	77.3%	85.0%	86.9%	87.5%	89.1%	89.9%	90.7%
GAAP Operating Margin (excl. restruct. & Other)	26.4%	73.6%	84.2%	77.8%	45.0%	67.7%	78.4%	81.0%	82.0%	84.0%	84.6%	85.5%
GAAP Net Income Margin	22.8%	62.4%	71.0%	65.5%	38.4%	57.8%	66.7%	68.2%	69.1%	70.8%	71.4%	72.2%
GAAP ROE %	17.2%	83.5%	86.5%	39.3%	37.1%	84.0%	120.1%	113.2%	102.1%	96.5%	85.7%	80.2%
Growth %												
Revenue Growth	48.9%	228.0%	107.0%	(18.4%)	20.6%	74.9%	60.0%	22.9%	14.1%	14.2%	7.6%	11.3%
Operating Profit Growth	836.4%	814.5%	136.7%	(24.6%)	66.2%	163.3%	85.3%	27.0%	15.4%	17.0%	8.5%	12.5%
Key Operating Metrics	FY2025	FY2026E	FY2027E	FY2028E	1QFY26	2QFY26	3QFY26E	4QFY26E	1QFY27E	2QFY27E	3QFY27E	4QFY27E
Total Revenue	37,378	122,598	253,776	207,112	13,643	23,860	38,183	46,911	53,529	61,154	65,829	73,264
DRAM Revenue	28,578	93,815	198,616	178,769	10,812	18,768	29,349	34,886	40,018	46,827	51,515	60,255
Gb Shipped	68,444	85,714	98,974	123,583	20,059	21,103	21,947	22,605	23,419	23,887	25,082	26,587
ASP (\$/Gb)	0.42	1.09	2.01	1.45	0.54	0.89	1.34	1.54	1.71	1.96	2.05	2.27
Actual Cost (\$/Gb)	0.22	0.20	0.22	0.21	0.20	0.19	0.20	0.21	0.22	0.22	0.22	0.22
NAND Revenue	8,502	28,410	54,780	27,963	2,743	4,997	8,740	11,930	13,416	14,231	14,219	12,914
GB Shipped — Trade	120,664	147,298	175,868	210,921	34,680	35,493	37,622	39,503	41,518	43,179	44,474	46,698
ASP (\$/GB) — Total	0.07	0.19	0.31	0.13	0.08	0.14	0.23	0.30	0.32	0.33	0.32	0.28
ASP (\$/GB) — Trade	0.07	0.19	0.31	0.13	0.08	0.14	0.23	0.30	0.32	0.33	0.32	0.28
Actual Cost (\$/GB) — Trade	0.06	0.06	0.05	0.04	0.06	0.06	0.06	0.06	0.05	0.05	0.05	0.05
NOR Revenue	200	200	200	200	50	50	50	50	50	50	50	50
Other Revenue	98	173	180	180	38	45	45	45	45	45	45	45
Operating Profit (excl. restruct. & Other)	9,870	90,265	213,614	161,149	6,138	16,161	29,947	38,019	43,885	51,350	55,702	62,677
Tangible BVPS	47.9	114.0	259.0	367.4	50.8	62.8	84.8	113.0	141.7	176.2	214.3	258.0
Company gross margins	39.8%	78.9%	87.8%	82.7%	56.0%	74.4%	82.7%	84.9%	86.1%	87.7%	88.1%	88.8%
DRAM gross margins	47.5%	81.8%	89.0%	85.5%	63.6%	79.1%	85.1%	86.2%	87.1%	88.8%	89.2%	90.2%
Total NAND gross margins	15.8%	70.4%	83.8%	66.3%	28.7%	58.0%	75.3%	81.6%	83.5%	84.6%	84.4%	82.5%
Trade NAND gross margins	15.8%	70.4%	83.8%	66.3%	28.7%	58.0%	75.3%	81.6%	83.5%	84.6%	84.4%	82.5%

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 34: Micron Balance Sheet and Cash Flow Statement

Figures in USD M Unless Otherwise Stated; Fiscal Year ends August 30

Balance Sheet	FY2025	FY2026E	FY2027E	FY2028E	1QFY26	2QFY26	3QFY26E	4QFY26E	1QFY27E	2QFY27E	3QFY27E	4QFY27E
Cash and equivalents	9,642	45,516	162,986	277,128	9,731	13,908	24,785	45,516	67,508	95,284	127,666	162,986
Restricted Cash	-	-	-	-	-	-	-	-	-	-	-	-
Receivables	9,265	34,041	53,164	23,276	10,184	17,314	27,112	34,041	39,275	44,376	46,741	53,164
Inventories	8,355	7,978	8,490	11,525	8,205	8,267	8,123	7,978	8,005	8,001	8,181	8,490
Other Current Assets	1,579	1,924	1,924	1,924	1,545	1,924	1,924	1,924	1,924	1,924	1,924	1,924
Total Current Assets	28,841	89,459	226,564	313,853	29,665	41,413	61,943	89,459	116,712	149,586	184,512	226,564
Property, Plant & Equipment	46,590	60,875	86,513	115,924	48,477	51,408	56,327	60,875	65,724	72,711	79,003	86,513
Intangible Assets & Goodwill	5,738	6,650	6,650	6,650	6,132	6,650	6,650	6,650	6,650	6,650	6,650	6,650
Long-Term marketable investments	1,629	2,038	2,039	2,043	1,697	2,038	2,037	2,038	2,039	2,039	2,039	2,039
Equity Method Investments	-	-	-	-	-	-	-	-	-	-	-	-
Other Non-Current Assets	-	-	-	-	-	-	-	-	-	-	-	-
Total Non-Current Assets	53,957	69,563	95,201	124,617	56,306	60,096	65,014	69,563	74,413	81,400	87,691	95,201
Total Assets	82,798	159,022	321,765	438,469	85,971	101,509	126,958	159,022	191,125	230,986	272,203	321,765
Accounts Payable	9,649	12,686	17,210	19,260	9,796	10,997	12,094	12,686	13,574	15,552	15,173	17,210
Debt	14,577	541	191	191	11,756	1,241	891	541	191	191	191	191
Other Liabilities	4,407	16,812	16,812	16,812	5,613	16,812	16,812	16,812	16,812	16,812	16,812	16,812
Total Liabilities	28,633	30,039	34,213	36,263	27,165	29,050	29,797	30,039	30,577	32,555	32,176	34,213
Minority Interest	-	-	-	-	-	-	-	-	-	-	-	-
Shareholder's Equity	54,165	128,983	287,553	402,206	58,806	72,459	97,161	128,983	160,548	198,431	240,028	287,553
Total Liabilities & Equity	82,798	159,022	321,765	438,469	85,971	101,509	126,958	159,022	191,125	230,986	272,203	321,765
Tangible BVPS	47.9	114.0	259.0	367.4	50.8	62.8	84.8	113.0	141.7	176.2	214.3	258.0
Cash Flow Statement	FY2025	FY2026E	FY2027E	FY2028E	1QFY26	2QFY26	3QFY26E	4QFY26E	1QFY27E	2QFY27E	3QFY27E	4QFY27E
Net Income	8,539	76,471	180,157	135,589	5,240	13,785	25,455	31,991	36,963	43,279	46,993	52,921
Depreciation & Amortization	8,352	9,755	13,362	18,589	2,212	2,286	2,501	2,757	2,951	3,153	3,459	3,800
Other	634	(17,958)	(15,113)	28,900	959	(4,168)	(8,555)	(6,193)	(4,374)	(3,119)	(2,924)	(4,695)
CF from Operations	17,525	68,269	178,406	183,077	8,411	11,903	19,400	28,555	35,540	43,313	47,528	52,026
Capex	(15,857)	(26,500)	(39,000)	(48,000)	(5,389)	(6,387)	(7,420)	(7,304)	(7,800)	(10,140)	(9,750)	(11,310)
Other	1,770	1,657	-	-	795	862	-	-	-	-	-	-
CF from Investing	(14,087)	(24,843)	(39,000)	(48,000)	(4,594)	(5,525)	(7,420)	(7,304)	(7,800)	(10,140)	(9,750)	(11,310)
CF from Financing	(850)	(7,535)	(21,937)	(20,935)	(3,745)	(2,167)	(1,104)	(519)	(5,748)	(5,397)	(5,396)	(5,396)
Beginning Cash	7,041	9,642	45,516	162,986	9,642	9,731	13,908	24,785	45,516	67,508	95,284	127,666
Net Cash Flow	2,601	35,874	117,470	114,142	89	4,177	10,877	20,731	21,992	27,776	32,381	35,320
Adjustments	-	-	-	-	-	-	-	-	-	-	-	-
Ending Cash	9,642	45,516	162,986	277,128	9,731	13,908	24,785	45,516	67,508	95,284	127,666	162,986
Free Cash Flow	1,668	41,769	139,406	135,077	3,022	5,516	11,980	21,251	27,740	33,173	37,778	40,716

Source: Company reports, Bernstein estimates and analysis

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**Samsung Electronics Co Ltd**

We value Samsung Electronics with 6.2x of forward 5Q-8Q EPS of KRW71,172 and arrive at price target of KRW 440,000 for its common shares, KRW374,000 for preferred shares, and US\$7,350 for GDR.

SK Hynix Inc

We value SK hynix with 6.2x of forward 5Q-8Q EPS of KRW536,158 and arrive at price target of KRW3,300,000.

Micron Technology Inc

We value Micron with 7.7x of forward 5Q-8Q EPS of US\$169.8 and arrive at price target of US\$1,300.

KIOXIA Holdings Corp

We value KIOXIA with 4.1x of 1-year forward EPS of ¥9,810 and arrive at price target of ¥40,000

MediaTek Inc

We set a NT\$4,380 1-year price target using a P/E multiple of 22x against our forward Q5-Q8 EPS estimate of NT\$199.

RISKS**Samsung Electronics Co Ltd**

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

SK Hynix Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

Micron Technology Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

KIOXIA Holdings Corp

Upside risk to our target price include 1) stronger than expected NAND demand growth, which likely would lead to better NAND pricing and hence company earnings, 2) favorable policy support from Japanese government such as capex subsidies, and 3) better than expected cost reduction

MediaTek Inc

There are a number of downside risks to our MediaTek price target: (1) Stronger competitive pressure from Qualcomm in 5G market (2) Slower 5G adoption increase, especially in China (3) Weaker smartphone demand globally (4) Slower diversification beyond smartphone (5) Downturn in the broad semiconductor market

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 500 Banks And Financial Services Index (BEBANKS) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 500 Insurance Index (BEINSUR) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Neutral (C-N): The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Underperform (C-UP): The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

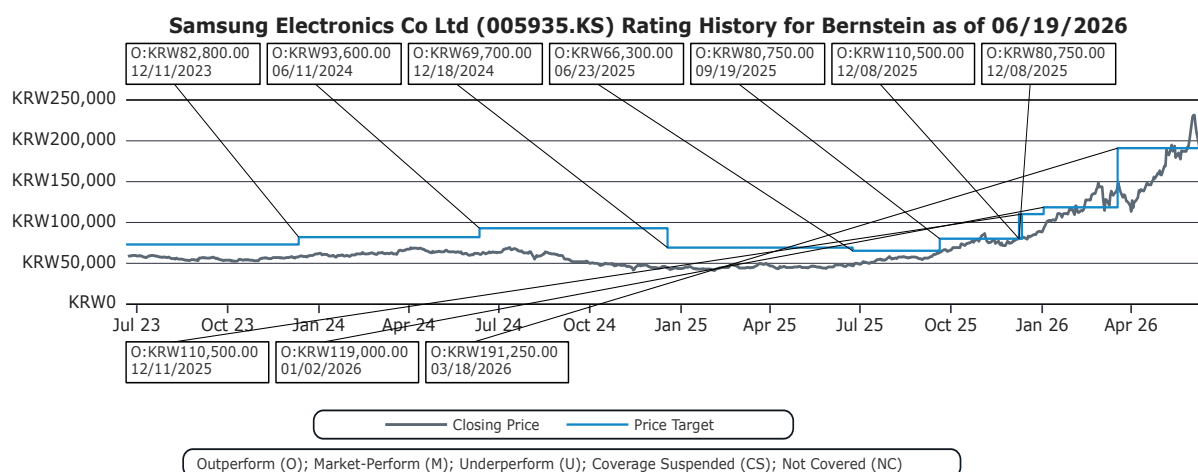
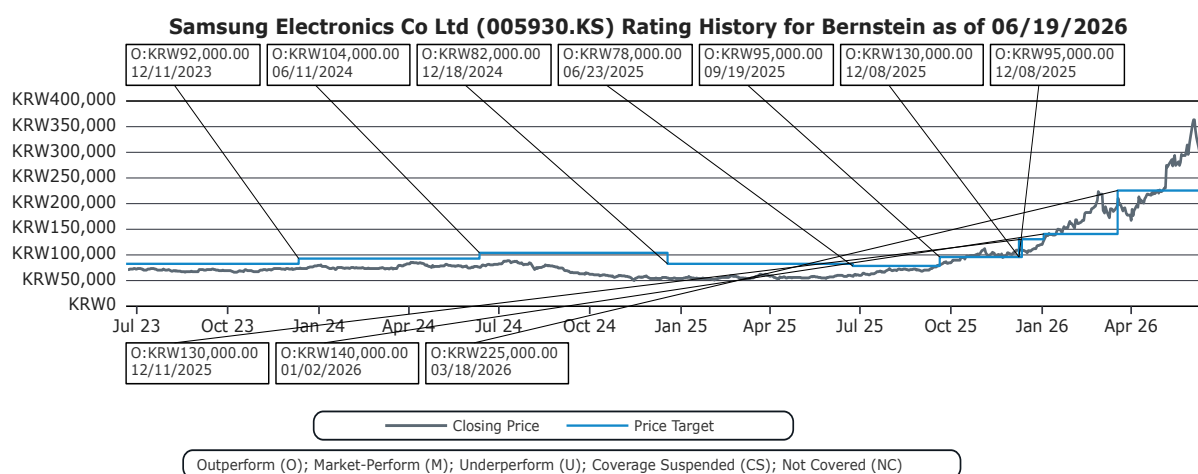
It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

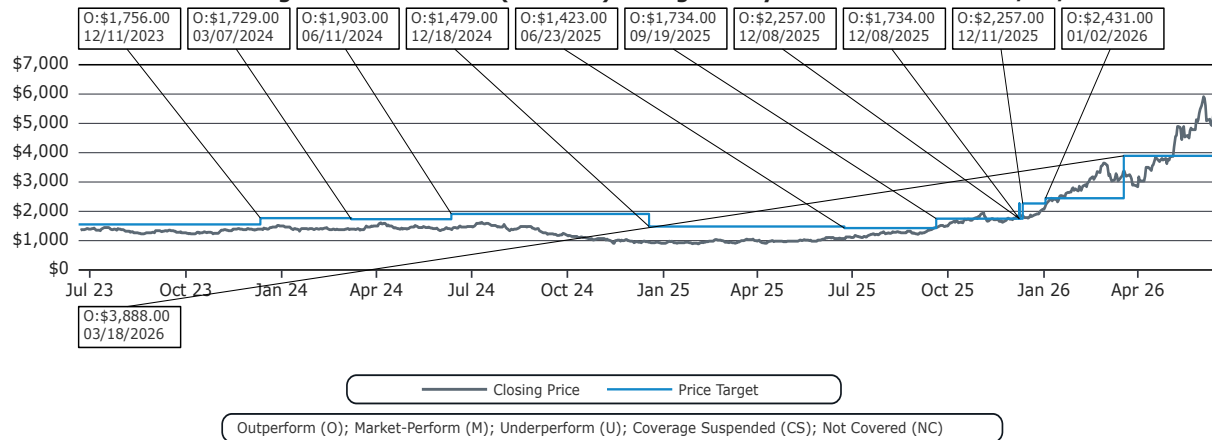
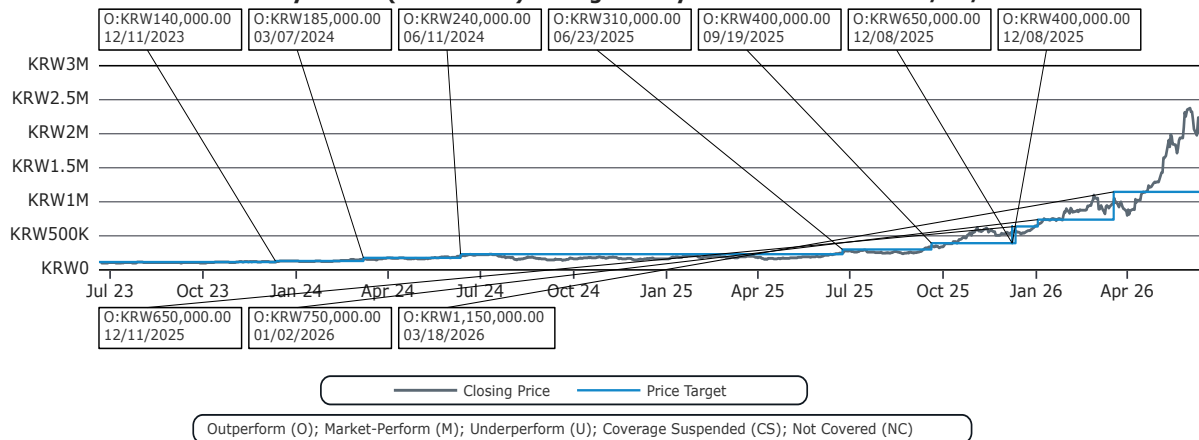
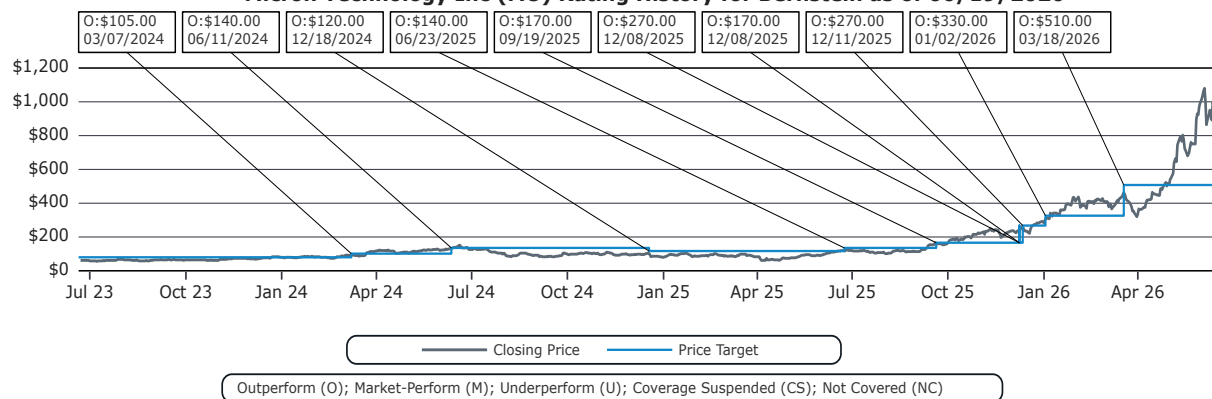
DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

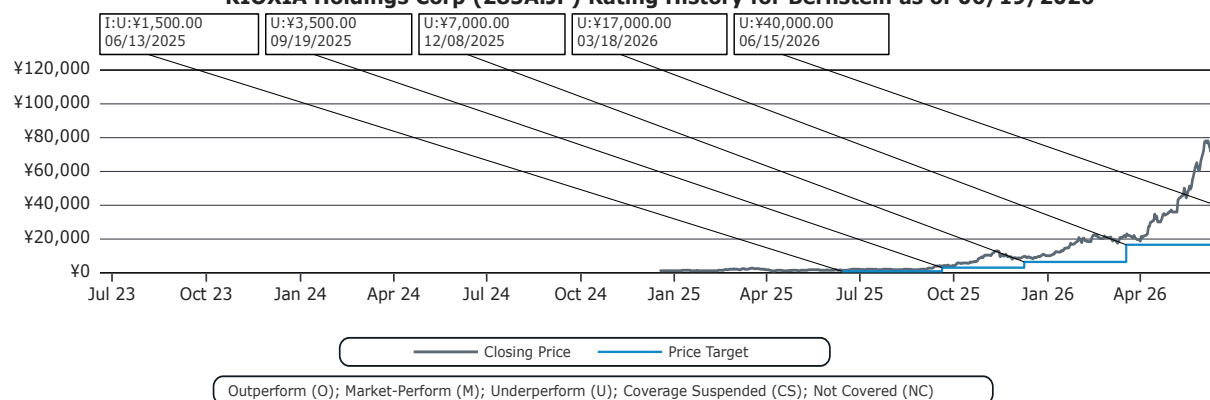
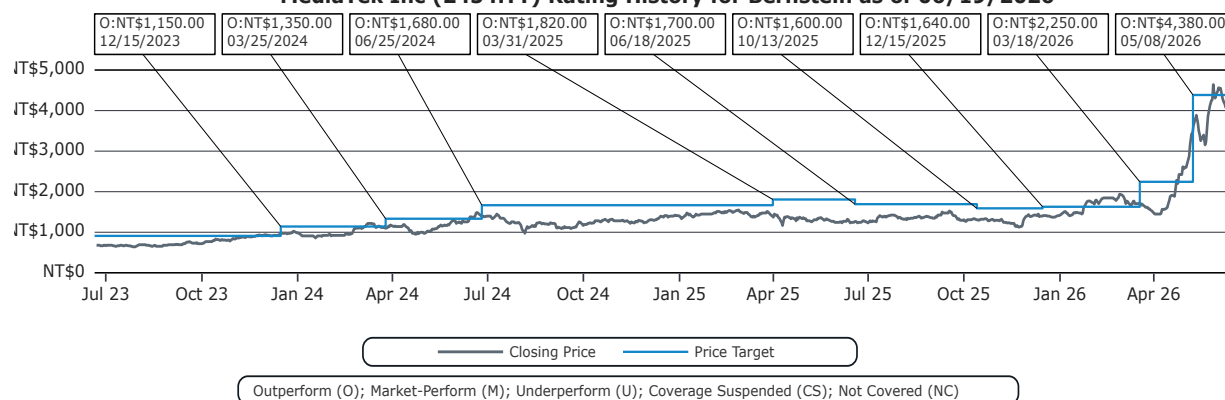
Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.1%	16.5%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	36.3%	17.8%
Underperform	SELL	12.6%	14.9%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of March 31, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY

Samsung Electronics Co Ltd (SMSN.LI) Rating History for Bernstein as of 06/19/2026**SK Hynix Inc (000660.KS) Rating History for Bernstein as of 06/19/2026****Micron Technology Inc (MU) Rating History for Bernstein as of 06/19/2026**

KIOXIA Holdings Corp (285A.JP) Rating History for Bernstein as of 06/19/2026**MediaTek Inc (2454.TT) Rating History for Bernstein as of 06/19/2026**

All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

CONFLICTS OF INTEREST

SG and/or its affiliates beneficially own 0.5% or more of [the total issued share capital/any class of common equity securities] with a net [long/short] position of: Micron Technology Inc.

SG and/or its affiliates beneficially own 1% or more of a class of common equity securities of the following company: KIOXIA Holdings Corp.

Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Samsung Electronics Co Ltd.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Samsung Electronics Co Ltd, SK Hynix Inc and KIOXIA Holdings Corp.

Certain affiliates of Bernstein act as market maker or liquidity provider in the debt securities of: SK Hynix Inc and KIOXIA Holdings Corp.

Certain affiliates of Bernstein act as market maker or liquidity provider in the equities securities of: Micron Technology Inc.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

- +1 Bernstein Institutional Services LLC; New York, New York, USA
- +44 Bernstein Autonomous LLP; London UK
- +212 Société Générale Africa Technologies & Services; Casablanca, Morocco
- +33 BSG France S.A.; Paris, France
- +34 BSG France S.A.; Madrid, Spain
- +41 Bernstein Autonomous LLP; Geneva, Switzerland
- +49 BSG France S.A.; Frankfurt, Germany
- +91 Sanford C. Bernstein (India) Private Limited; Mumbai, India
- +852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China
- +65 Sanford C. Bernstein (Singapore) Private Limited; Singapore
- +81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for "Bernstein" and "Autonomous" research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the "Autonomous" and "Bernstein" brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.
- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to "Autonomous" branded products (including certain Sales materials) please visit:

www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation ("MAR") and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission ("SEC") and a member of the U.S. Financial Industry Regulatory Authority, Inc. ("FINRA") is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as "permitted client" as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. ("BTG") is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Financial Promotion Order"), (ii) are persons falling within Article 49(2)(a) to (d) ("high net worth companies, unincorporated associations, etc.") of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as "relevant persons"). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (<https://www.sfc.hk/publicregWeb/corp/AXC846/details>). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore ("SFA"). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration

No. 20213710W and located at 8 Marina Boulevard, #12-01, Marina Bay Financial Centre, Singapore 018981, +65-6326-7000.

To our readers in the People's Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the "PRC") in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Investment Management Association of Japan. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. ("Daiwa"), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;
- making a market for a financial product; and
- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.
- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives

- do not have any financial interest in the subject company
- do not have actual/beneficial ownership of one percent or more in securities of the subject company;
- is not engaged in any investment banking activities for Indian companies, as such;
- have not managed or co-managed a public offering in the past twelve months for any Indian companies;
- have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
- have not received compensation for brokerage services from the subject company in the past twelve months;
- have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and
- do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.
- do not have any conflict of interest in the subject company as of the date of this report.
- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.
- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.
- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com
- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at [Sanford C. Bernstein \(India\) Private Limited](https://bernsteinresearch.in/) (<https://bernsteinresearch.in/>) for your reference.
- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park

Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are "eligible counterparties", "professional clients", "institutional investors" and/or "professional investors" as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein's express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein's express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein's research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)' current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or

subscribe for shares, or to induce engagement in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients' home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.

No part of this material may be reproduced, distributed or transmitted or otherwise made available without prior consent of the entities noted herein. Copyright Bernstein Institutional Services LLC Bernstein Autonomous LLP, BSG France S.A., Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited and Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社). All rights reserved. The trademarks and service marks contained herein are the property of their respective owners. Any unauthorized use or disclosure is strictly prohibited. The entities noted herein may pursue legal action if the unauthorized use results in any defamation and/or reputational risk to the entities noted herein and research published under the Bernstein and Autonomous brands.